RF DEFINED AT TWO VERTICES OF A DISTINGUISHED TRIANGLE IS DEFINED AT THE THIRD

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4908-5016, following the statement at 4895-4906.

Editorial note. The statement's closing 'Similarly for LF' is not executed in the proof: mirror of the RF argument (left derived functor via the opposite triangulated category), recorded as a mirror-level omission.

Allowed standard prerequisites (editorial).

- Induced morphism RF(f). lemma-derived-functor, cited at proof line 4911; complete proof at source lines (queued, pb 746), same chapter. Supplies the induced morphism RF(f) : A -> B between chosen values.
- Reduction lemma for definedness. lemma-derived-inverts, cited at proof line 4965; short source lemma (pb 32) in the same chapter. Supplies that it suffices to prove RF defined at Z' with the arrow gamma inducing the isomorphism C -> RF(Z').
- Category of triangles over a fixed triangle. lemma-limit-triangles, cited at proof lines 4975 and 4986; its span 1812-1952 partially overlaps published range 316 and its proof in the chapter is complete (cofinality statement included). Supplies the category I of triangles over (X', Y', Z') and the cofinality of I -> Z'/S.
- Exactness of filtered colimits. algebra-lemma-directed-colimit-exact, cross-chapter citation at proof line 5005 (tags mechanism). Supplies that the left column of the Mor diagram is an exact sequence.
- Five lemma. homology-lemma-five-lemma, cross-chapter citation at proof line 5007 (tags mechanism). Supplies that colim Mor(W, F(Z")) -> Mor(W, C) is bijective.
- Characterization of essentially constant ind-objects. categories-lemma-characterize-essentially-constant-ind part (4), cross-chapter citation at proof line 5014 (tags mechanism). Supplies that the bijection for all W means F(Z") is essentially constant over I with value C.

Source reference. Original reference 00DB: focused-seeds/geometry/S9-der-2-out-of-3-defined/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05PU: focused-seeds/geometry/S9-der-2-out-of-3-defined/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Source reference. Original reference 05PY: focused-seeds/geometry/S9-der-2-out-of-3-defined/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A1S: focused-seeds/geometry/S9-der-2-out-of-3-defined/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05QB: focused-seeds/geometry/S9-der-2-out-of-3-defined/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SA: sources/stacks-derived/pinned-original-source.tex, source lines 4789–4789. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SB: sources/stacks-derived/pinned-original-source.tex, source lines 4855–4855. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05R9: sources/stacks-derived/pinned-original-source.tex, source lines 1813–1813. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05S8: sources/stacks-derived/pinned-original-source.tex, source lines 4736–4736. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Assumptions and notation as in Situation 05S8. Let (X, Y, Z, f, g, h) be a distinguished triangle of $\mathcal{D}$. If RF is defined at two out of three of X, Y, Z, then it is defined at the third. Moreover, in this case*

$$(RF(X), RF(Y), RF(Z), RF(f), RF(g), RF(h))$$

is a distinguished triangle in $\mathcal{D}'$. Similarly for LF.

Proof. Say RF is defined at X, Y with values A, B. Let $RF(f) : A \to B$ be the induced morphism, see Lemma 05SA. We may choose a distinguished triangle $(A, B, C, RF(f), b, c)$ in $\mathcal{D}'$. We claim that C is a value of RF at Z.

To see this pick $s : X \to X'$ in S such that there exists a morphism $\alpha : A \to F(X')$ as in Categories, Definition 05PU. We may choose a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{s} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xrightarrow{s'} & Y' \end{array}$$

with $s' \in S$ by MS2. Using that Y/S is filtered we can (after replacing s' by some $s'' : Y \to Y''$ in S) assume that there exists a morphism $\beta : B \to F(Y')$ as in Categories, Definition 05PU. Picture

$$\begin{array}{ccccc} A & \xrightarrow{\alpha} & F(X') & \longrightarrow & A \\ {\scriptstyle RF(f)}\downarrow & & \downarrow{\scriptstyle F(f')} & & \downarrow{\scriptstyle RF(f)} \\ B & \xrightarrow{\beta} & F(Y') & \longrightarrow & B \end{array}$$

It may not be true that the left square commutes, but the outer and right squares commute. The assumption that the ind-object $\{F(Y')\}_{s':Y' \to Y}$ is essentially constant means that there exists a $s'' : Y \to Y''$ in S and a morphism $h : Y' \to Y''$ such that $s'' = h \circ s'$ and such that $F(h)$ equal to $F(Y') \to B \to F(Y') \to F(Y'')$. Hence after replacing Y' by Y'' and β by $F(h) \circ \beta$ the diagram will commute (by direct computation with arrows).

Using MS6 choose a morphism of triangles

$$(s, s', s'') : (X, Y, Z, f, g, h) \longrightarrow (X', Y', Z', f', g', h')$$

with $s'' \in S$. By TR3 choose a morphism of triangles

$$(\alpha, \beta, \gamma) : (A, B, C, RF(f), b, c) \longrightarrow (F(X'), F(Y'), F(Z'), F(f'), F(g'), F(h'))$$

By Lemma 05SB it suffices to prove that $RF(Z')$ is defined and that the arrow $\gamma : C \to F(Z')$ induces an isomorphism $C \to RF(Z')$. Namely, then we will get an isomorphism

$$(A, B, C, RF(f), b, c) \longrightarrow (RF(X'), RF(Y'), RF(Z'), RF(f'), RF(g'), RF(h'))$$

of triangles and by TR1 we conclude that the target is a distinguished triangle. Consider the category $\mathcal{I}$ of Lemma 05R9 of triangles

$$\mathcal{I} = \{(t, t', t'') : (X', Y', Z', f', g', h') \to (X'', Y'', Z'', f'', g'', h'') \mid (t, t', t'') \in S\}$$

To show that the system $F(Z'')$ is essentially constant over the category Z'/S is equivalent to showing that the system of $F(Z'')$ is essentially constant over $\mathcal{I}$ because $\mathcal{I} \to Z'/S$ is cofinal, see Categories, Lemma 0A1S (cofinality is proven in

Lemma 05R9). For any object W in $\mathcal{D}'$ we consider the diagram

$$\begin{array}{ccc}
\operatorname{colim}_{\mathcal{I}} \operatorname{Mor}_{\mathcal{D}'}(W, F(X'')) & \longleftarrow & \operatorname{Mor}_{\mathcal{D}'}(W, A) \\
\downarrow & & \downarrow \\
\operatorname{colim}_{\mathcal{I}} \operatorname{Mor}_{\mathcal{D}'}(W, F(Y'')) & \longleftarrow & \operatorname{Mor}_{\mathcal{D}'}(W, B) \\
\downarrow & & \downarrow \\
\operatorname{colim}_{\mathcal{I}} \operatorname{Mor}_{\mathcal{D}'}(W, F(Z'')) & \longleftarrow & \operatorname{Mor}_{\mathcal{D}'}(W, C) \\
\downarrow & & \downarrow \\
\operatorname{colim}_{\mathcal{I}} \operatorname{Mor}_{\mathcal{D}'}(W, F(X''[1])) & \longleftarrow & \operatorname{Mor}_{\mathcal{D}'}(W, A[1]) \\
\downarrow & & \downarrow \\
\operatorname{colim}_{\mathcal{I}} \operatorname{Mor}_{\mathcal{D}'}(W, F(Y''[1])) & \longleftarrow & \operatorname{Mor}_{\mathcal{D}'}(W, B[1])
\end{array}$$

where the horizontal arrows are given by composing with (α, β, γ). Since filtered colimits are exact (Algebra, Lemma 00DB) the left column is an exact sequence. Thus the 5 lemma (Homology, Lemma 05QB) tells us the map

$$\operatorname{colim}_{\mathcal{I}} \operatorname{Mor}_{\mathcal{D}'}(W, F(Z'')) \longrightarrow \operatorname{Mor}_{\mathcal{D}'}(W, C)$$

is bijective. We conclude that $F(Z'')$ is essentially constant over $\mathcal{I}$ with value C by part (4) of Categories, Lemma 05PY. $\square$